

Question ID: b544a348

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$5x + 3y = 38$$
$$x + 3y = 10$$

In the solution (x, y) to the system of equations above, what is the value of x ?

Rationale

The correct answer is 7. Subtracting the second equation from the first equation eliminates the variable y .

$$\begin{array}{rcl} 5x + 3y & = & 38 \\ -(x + 3y & = & 10) \\ \hline 4x & = & 28 \end{array}$$

Dividing both sides of the resulting equation by 4 yields $x = 7$.